

Foundations of Stacking ensemble

[Stacking ensemble]

1.0 Content Block

Land cover mapping Land cover mapping is one of the major applications of Earth observation satellite sensors, using remote sensing and geospatial data, to identify the materials and objects which are located on the surface of target areas. Generally, the classes of target materials include roads, buildings, rivers, lakes, and vegetation. Some different ensemble learning approaches based on artificial neural networks, kernel principal component analysis (KPCA), decision trees with boosting, random forest and automatic design of multiple classifier systems, are proposed to efficiently identify land cover objects.

2.0 Content Block

In statistics and machine learning, ensemble methods use multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone. Unlike a statistical ensemble in statistical mechanics, which is usually infinite, a machine learning ensemble consists of only a concrete finite set of alternative models, but typically allows for much more flexible structure to exist among those alternatives.

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Cross-Validation Selection can be summed up as: "try them all with the training set, and pick the one that works best". Gating is a generalization of Cross-Validation Selection. It involves training another learning model to decide which of the models in the bucket is best-suited to solve the problem. Often, a perceptron is used for the gating model. It can be used to pick the "best" model, or it can be used to give a linear weight to the predictions from each model in the bucket. When a bucket of models is used with a large set of problems, it may be desirable to avoid training some of the models that take a long time to train. Landmark learning is a meta-learning approach that seeks to solve this problem. It involves training only the fast (but imprecise) algorithms in the bucket, and then using the performance of these algorithms to help determine which slow (but accurate) algorithm is most likely to do best.

4.0 Content Block

Boosting involves training successive models by emphasizing training data mis-classified by previously learned models. Initially, all data (D1) has equal weight and is used to learn a base model M1. The examples mis-classified by M1 are assigned a weight greater than correctly classified examples. This boosted data (D2) is used to train a second base model M2, and so on. Inference is done by voting. In some cases, boosting has yielded better accuracy than bagging, but tends to over-fit more. The most common implementation of boosting is Adaboost, but some newer algorithms are reported to achieve better results.

5.0 Content Block

Overview Supervised learning algorithms search through a hypothesis space to find a suitable hypothesis that will make good predictions with a particular problem. Even if this space contains hypotheses that are very well-suited for a particular problem, it may be very difficult to find a good one. Ensembles combine multiple hypotheses to form one which should be theoretically better. Ensemble learning trains two or more machine learning algorithms on a specific classification or regression task. The algorithms within the ensemble model are generally referred as "base models", "base learners", or "weak learners" in literature. These base models can be constructed using a single modelling algorithm, or several different algorithms. The idea is to train a diverse set of weak models on the same modelling task, such that the outputs of each

weak learner have poor predictive ability (i.e., high bias), and among all weak learners, the outcome and error values exhibit high variance. Fundamentally, an ensemble learning model trains at least two high-bias (weak) and high-variance (diverse) models to be combined into a better-performing model. The set of weak models — which would not produce satisfactory predictive results individually — are combined or averaged to produce a single, high performing, accurate, and low-variance model to fit the task as required. Ensemble learning typically refers to bagging (bootstrap aggregating), boosting or stacking/blending techniques to induce high variance among the base models. Bagging creates diversity by generating random samples from the training observations and fitting the same model to each different sample — also known as homogeneous parallel ensembles. Boosting follows an iterative process by sequentially training each base model on the up-weighted errors of the previous base model, producing an additive model to reduce the final model errors — also known as sequential ensemble learning. Stacking or blending consists of different base models, each trained independently (i.e. diverse/high variance) to be combined into the ensemble model — producing a heterogeneous parallel ensemble. Common applications of ensemble learning include random forests (an extension of bagging), Boosted Tree models, and Gradient Boosted Tree Models. Models in applications of stacking are generally more task-specific — such as combining clustering techniques with other parametric and/or non-parametric techniques. Evaluating the prediction of an ensemble typically requires more computation than evaluating the prediction of a single model. In one sense, ensemble learning may be thought of as a way to compensate for poor learning algorithms by performing a lot of extra computation. On the other hand, the alternative is to do a lot more learning with one non-ensemble model. An ensemble may be more efficient at improving overall accuracy for the same increase in compute, storage, or communication resources by using that increase on two or more methods, than would have been improved by increasing resource use for a single method. Fast algorithms such as decision trees are commonly used in ensemble methods (e.g., random forests), although slower algorithms can benefit from ensemble techniques as well. By analogy, ensemble techniques have been used also in unsupervised learning scenarios, for example in consensus clustering or in anomaly detection.

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$$y = \underset{c_j \in C}{\operatorname{argmax}} \sum_{h_i \in H} P(c_j | h_i) P(h_i | T) \quad \{\displaystyle y = \{\underset{c_j \in C}{\operatorname{argmax}}\} \sum_{h_i \in H} P(c_j | h_i) P(h_i | T)\}$$

7.0 Content Block

Stacking Stacking (sometimes called stacked generalization) involves training a model to combine the predictions of several other learning algorithms. First, all of the other algorithms are trained using the available data, then a combiner algorithm (final estimator) is trained to make a final prediction using all the predictions of the other algorithms (base estimators) as additional inputs or using cross-validated predictions from the base estimators which can prevent overfitting. If an arbitrary combiner algorithm is used, then stacking can theoretically represent any of the ensemble techniques described in this article, although, in practice, a logistic regression model is often used as the combiner. Stacking typically yields performance better than any single one of the trained models. It has been successfully used on both supervised learning tasks (regression, classification and distance learning) and unsupervised learning (density estimation). It has also been used to estimate bagging's error rate. It has been reported to out-perform Bayesian model-averaging. The two top-performers in the Netflix competition utilized blending, which may be considered a form of stacking. Bayesian predictive stacking generalizes the idea of stacking from the statistical estimation literature to the combination of posterior predictive distributions. These ideas have also been developed and investigated for Gaussian process models, especially for spatial data analysis and can be used to construct transfer learning frameworks for massive spatial data sets.